

Cardiff School of Technologies

Assessment Brief

Module Code

DAT5003

Module Title

Applied Data Science

Academic Year

24-25

Semester

2

Module Leader email

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Content

Assessment Details	2
Submission Details	3
Assessment Criteria	3
Further Information	5
Who can answer questions about my assessment?	5
Referencing	5
Submission problems	5
Unfair academic practice	5
How is my work graded?	6

Assessment Details

Assessment title	Abr.	Weighting
Project Report	WRIT1	70%

Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.

Task/assessment brief:

Assessment Requirements / Tasks (include all guidance notes)

CONTEXT

For this assignment, you are required to select one of the following datasets:

1. Online news popularity data (<https://archive.ics.uci.edu/dataset/332/online+news+popularity>)
2. Online retail (<https://archive.ics.uci.edu/dataset/352/online+retail>)
3. Air quality (<https://archive.ics.uci.edu/dataset/360/air+quality>)
4. Airbnb (<https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata>)

The data files and their respective URLs are also available on Moodle under the assessment submission point.

This assignment is designed to assess your ability to formulate relevant and insightful analytic questions based on your chosen dataset. You will need to process the data using key steps of (big) data analytics, including data pre-processing, analysis, and the preparation of an analytical report. Your analysis should address a significant issue that is valuable for stakeholders (e.g., policymakers, businesses) to gain meaningful insights, rather than a trivial topic.

Task 1 – Analytic question

In this task, choose an analytic question. For your analysis to be compelling it must address a substantive issue useful for the stakeholders (for example, policymakers, businesses) to draw insight rather than a trivial one. Hence you must consider these aspects before choosing a question. An example question could be:

- Is there any noticeable difference of traffic among different areas/cities and what could be the possible reason(s) for it? – **Relevant data: Airbnb**
- What factors account for difference of pricing between different areas/cities and could Airbnb use listing descriptions (or part thereof) to determine optimal listing price (so they may be able to suggest if a property is listed above/below appropriate price)? – **Relevant data: Airbnb**
- What clusters of products are commonly purchased together, and how can these insights be used for cross-selling and bundling strategies? – **Relevant data: Online retail**
- Which products are likely to be returned, based on factors such as quantity purchased, price, and purchase frequency? – **Relevant data: Online retail**

You need to discuss your analytic question with the Module Leader before the end of week 6 of teaching and collect their feedback to refine your question if necessary.

Deliverable

The analytic questions for Task 1 should be submitted in a .ipynb file. In addition to the questions, you should provide a concise explanation of the rationale behind your chosen question and the expected impact of answering it.

Task 2 – Data Analysis

In this task, use your data analytics skills to answer the question posed in Task 1. Depending upon your chosen question, you will typically have to perform exploratory data analysis (EDA), data pre-processing, statistics-based analysis, visualisation, and optional predictive modelling (if your case requires it). To handle the data, you might use suitable database but that is entirely optional and will not have any impact upon your grade.

Deliverable

The solution for Task 2 should be included in the same .ipynb file as Task 1. You must explain each step of the data analysis process and provide the corresponding working code. The .ipynb file should run smoothly on any standard machine without errors, and all results must be reproducible.

Task 3 – Analytical report

In this task, you are expected to prepare an analytical report, which can easily be interpreted by a non-technical audience like executive board members or policymakers. There is no fixed specification for this report but in general it must contain an adequate number of visualisation charts/graphs with lucid description. **The word limit for this task is 280.**

Deliverable

The solution for Task 3 should be submitted as a PDF file.

Word count (or equivalent):

5600 words

This is a reflection of the effort required for the assessment per student. Word counts will include any text, tables, subtitles and citations. Reference lists are excluded from the word count.

Academic or technical terms explained:

EDA – it is the process of examining and summarising datasets to uncover patterns, trends, and relationships while identifying anomalies or missing data. It involves the use of visualisations, statistical measures, and descriptive techniques to understand the data's structure and inform further analysis. EDA is a critical first step in data analysis, providing insights that guide modelling and decision-making

Data pre-processing – it is the process of preparing raw data for analysis by cleaning, transforming, and organising it into a usable format. It involves tasks such as handling missing values, removing duplicates, normalizing or scaling features, and encoding categorical variables. Effective data pre-processing ensures the quality and consistency of data, which is essential for accurate and reliable machine learning or statistical analysis.

Statistics-based analysis: it refers to the use of statistical methods and techniques to summarise, interpret, and infer insights from data. It involves applying descriptive and inferential statistics to identify trends, relationships, and patterns, providing a foundation for decision-making and hypothesis testing.

Predictive modelling – it is the process of using statistical or machine learning techniques to create a model that forecasts future outcomes based on historical data. It involves identifying patterns in the data to make predictions about unknown or future events, such as customer behaviour, sales trends, or risk assessments.

Key Bloom elements:

Application – Application of Python-based tools such as Pandas, scikit-learn to arrange data, pre-process data, and prepare data for analysis.

Analysis – Various stages of data analysis such as EDA, regression, classification, clustering etc. to answer the chosen analytic question.

Synthesis – Aggregating the major components of data analysis to prepare report for non-technical audience.

Artificial Intelligence Models – Guidance for this assessment:

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has provided some resources to support you in its appropriate usage:

- [Library Services AI Hub](#)
- [Student Guide to AI and Assessment](#)
- [Code of Conduct for Students on the use of AI](#)
- [Cite Them Right resource on citing materials relating to AI \(if permitted\)](#)

As per the academic regulations ([Academic Handbook Ah1_08](#)), in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including that which is AI generated. Upon submission of work, you will be asked to confirm the following statement:

I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.
- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment. Remember that in all cases you must submit work that is your own, acknowledging any part of it that has been provided by another source.

NO USE OF GENERATIVE AI EXPECTED • Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately. • AI tools for checking spelling, grammar and referencing may be used.	
AI ACKNOWLEDGED • You can use AI tools to learn about your topic, as part of your study, or in preparing initial guidance on assignments (e.g. headline structure, suggestions for inclusion of topics). • Any materials that you have sourced from AI should be rewritten or reconfigured and integrated into your own work and referenced	

appropriately. It is recommended that this is confirmed by a relevant academic source.

- Any support gained from AI should be acknowledged in a statement at the end of the assignment, making clear what the support was, and how you used it and developed it for your own work. Example statements are available in the Student Code of Conduct [\[link\]](#).

AI EMBEDDED

- Use of AI is an integral and expected part of the assessment.
- The explicit inclusion of AI within the assessment means that instructions on the expected use will be part of the assessment brief.
- Your assessment brief will describe how you should acknowledge the way in which you used AI tools.

Submission Details

Submission Deadline:	Check Moodle for submission dates	Estimated Feedback Return Date	This will normally be 20 working days after initial submission.
Submission Time:	By 4.00pm on the deadline day.		
Moodle/Turnitin:	Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.		
File Format:	The assessment must be submitted through the Turnitin submission point in Moodle. You must submit a .ipynb file (Task 1 and Task 2) and a PDF (Task 3) file. Your assessment should be titled with your: student ID number, module code and assessment ID, e.g. st12345678 DAT5003 WRIT1		
Feedback	Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the guidance on how to access your feedback. All marks are provisional and are subject to quality assurance processes and confirmation at the programme Examination Board.		

Assessment Criteria

Learning outcomes assessed

Learning Outcomes	Assessment Criteria
[LO1] Evaluate different data formats	Task 1 – Interesting analytic question (LO1, LO5)
[LO2] Analyse datasets through various statistical and computational techniques.	Task 2 – Data analysis (LO1, LO2, LO3, LO4)
[LO3] Discuss and apply solutions for processing datasets and solving complex problems in various domains.	Task 3 – Analytic report (LO3, LO5, LO6)
[LO4] Apply flexible and creative techniques to build real world applications.	
[LO5] Evaluate security risks with privacy related ethics.	
[LO6] Reflect on their own personal development within the module.	

Marks allocation

WRIT1	100%
Task 1 – Interesting analytics question	10%
Substantive question (LO1)	5%
Rationale (LO1, LO5)	3%
Expected impact (LO5)	2%
Task 2 – Data analysis	70%
Exploratory data analysis (LO1)	10%
Data pre-processing (such as, data cleaning, transformation etc. as appropriate) (LO1, LO2)	20%
Statistics -based (descriptive, prescriptive, predictive) data analytics (LO3, LO4)	30%
Visualisations of result (LO3, LO4)	10%
Task 3 – Analytical report (LO3, LO5, LO6)	20%

Other skills/attributes developed.

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

Ethical	Ethical issues surrounding core/non-core data science are considered as part of the design and implementation.
Digital	A low-scale project will be produced that similarly match the requirements encountered in real-problems.

Global	The market for the case studies must be considered and whether there are any cross-cultural issues which emerge.
Entrepreneurial	Professional data scientist techniques will be employed in a small cross disciplinary team, like that encountered in a start-up or idea-project.

Marking/Assessment Criteria

L5	80%-100%	70%-79%	60%-69%	50%-59%	40%-49%	30-39%	1-29%
Task 1 – Interesting analytic question	The work presents a specific and substantive analytic question that is both theoretically interesting and of significant practical importance. It demonstrates an excellent understanding of the given data, with error-free content that reflects considerable time and effort in preparing the question. The rationale for the question is well-founded, and there is a clear understanding of its expected business impact.	The work presents a substantive and specific analytic question that is theoretically interesting and has some practical importance. It demonstrates a high level of understanding of the given data, with only minor errors, reflecting a good amount of time and effort spent on preparation. The rationale for the question is very good, with a clear understanding of its expected business impact.	The work presents a clear analytic question that is relevant and demonstrates a solid understanding of the given data. It contains some minor errors, indicating adequate effort and time spent on question preparation. The rationale for the question is reasonable, and there is a general understanding of its potential business impact.	The work presents a focused analytic question with a clear understanding of the given data. It contains a few errors, which are appropriate given the time and effort spent on preparation. The rationale for the question is clear, with an understanding of its expected business impact.	The work presents a substantive but overly ambitious analytic question with reasonable evidence of understanding the selected topic. It contains some errors, but there is clear evidence of effort in the question preparation. The rationale for the question is adequate, with a clear understanding of its expected business impact.	The work presents an unfocused and unspecific analytic question, with some errors. While there is evidence of some effort in question preparation, the rationale is weak, and the understanding of the expected business impact is unclear.	The work presents an obvious and trivial research question with no practical importance, showing no understanding of the given data. It contains significant errors, reflecting minimal effort in question preparation. There is no rationale for the question, and the expected business impact is not understood.

Task 2 – Data analysis							
	Provides a comprehensive and error-free solution to the chosen question, with an in-depth and fully compliant design. Data analytics are executed independently and presented systematically, demonstrating clarity, professionalism, and adherence to best practices.	Delivers a thorough and accurate solution to the chosen question, with a well-structured design that largely meets required standards. Data analytics are executed effectively and presented in a clear and logical manner, demonstrating professionalism and good practice, with minor areas for improvement.	Provides a clear and mostly accurate solution to the chosen question, with a design that meets required standards but may lack depth or have minor inconsistencies. Data analytics are carried out and presented adequately, though improvements in clarity, structure, or adherence to best practices are needed.	Offers a basic and generally accurate solution to the chosen question, though the design may lack detail or contain some errors. Data analytics are attempted and partially presented, but there are noticeable gaps in clarity, structure, or alignment with required standards and best practices.	Provides a limited solution to the chosen question, with significant gaps in accuracy or completeness. The design may lack coherence or fail to meet required standards. Data analytics are attempted but are incomplete or poorly presented, with noticeable issues in clarity, structure, and adherence to best practices.	Demonstrates an incomplete or flawed solution to the chosen question, with limited evidence of understanding. The design is underdeveloped and fails to meet required standards. Data analytics are minimally attempted, with significant inaccuracies or omissions, and lack clarity, structure, and alignment with best practices.	Shows little to no evidence of understanding or ability to address the chosen question. The solution is incomplete or absent, with a design that does not meet required standards. Data analytics are either not attempted or are incorrect, with no clarity, structure, or adherence to best practices.
Task 3 - Analytical report							
	Excellent structured analytical report with professional presentation. Writing is clear, precise, and free from errors, demonstrating a professional and informative style.	Well-structured analytical report with a professional presentation. Writing is clear and informative, with minimal errors. Demonstrates strong communication.	Analytical report is adequately structured and presented. Writing is generally clear and informative, though minor errors or lapses in style may be present. Communication	Report structure and presentation are basic, with some clarity issues. Writing is generally understandable, but may contain noticeable errors or inconsistencies.	Report structure and presentation are lacking, with several areas of confusion. Writing contains frequent errors that impact clarity. Communication	Report structure is unclear and poorly organised, with significant issues in presentation. Writing contains frequent errors, affecting readability and clarity. Communication	Report lacks structure and organisation, with minimal effort in presentation. Writing is unclear, contains numerous errors, and is difficult to follow. Communication

	Outstanding communication of knowledge.	tion of knowledge and an effective writing style.	ation of knowledge is effective but could be more refined.	cies. Communication of knowledge is adequate, though more clarity and refinement are needed.	ation of knowledge is limited, and further improvement in clarity and accuracy is required.	ation of knowledge is minimal, and substantial improvement is needed in both content and presentation.	ation of knowledge is insufficient, and significant revision is needed in both content and clarity.
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Further Information on assessment, referencing and grading can be found in the Module Handbook (on Moodle)

An abstract graphic design featuring a dark blue background with a complex, light blue circuit-like pattern. The pattern consists of numerous thin, parallel lines that branch out and connect to various circular nodes of different sizes. The lines and nodes are arranged in a way that suggests a network or a data flow, with some lines running vertically and others branching out horizontally or diagonally. The overall effect is a sense of connectivity and technology.

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